Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

Reviewed example = a checked experiment; GenAge annotation = catalogue evidence only.

All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C007 **TRX1** 5 genes / 4 species 5 direct pairs	**Y1 TRX1** GenAge 1286 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20421943, 20883526	**W1 trx-1** GenAge 31 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 16324156	**F1 Trx-2** GenAge 685 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 17567437 **F2 TrxT** GenAge 686 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 17301052	**M1 Txn1** GenAge 691 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 12230882, 21873593
Direct pairs: W1–M1 (m:1); W1–Y1 (n:m); F2–M1 (1:1); F1–Y1 (n:m); M1–Y1 (1:m)				
C064 **TSA1** 4 genes / 4 species 3 direct pairs	**Y1 TSA1** GenAge 2109 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 27264606	**W1 prdx-2** GenAge 516 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 16673436	**F1 Jafrac1** GenAge 856 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 19720829, 20976250	**M1 Prdx1** GenAge 518 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 12891360
Direct pairs: W1–Y1 (1:m); F1–Y1 (n:m); M1–Y1 (n:m)				
C077 **AAC3** 4 genes / 3 species 5 direct pairs	**Y1 AAC3** GenAge 1553 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18340043 **Y2 PET9** GenAge 1595 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18809618	**W1 ant-1.1** GenAge 666 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17608836	No gene in this candidate network	**M1 Slc25a4** GenAge 615 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 10220377
Direct pairs: W1–M1 (m:n); W1–Y1 (n:m); W1–Y2 (n:m); M1–Y1 (n:m); M1–Y2 (n:m)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.